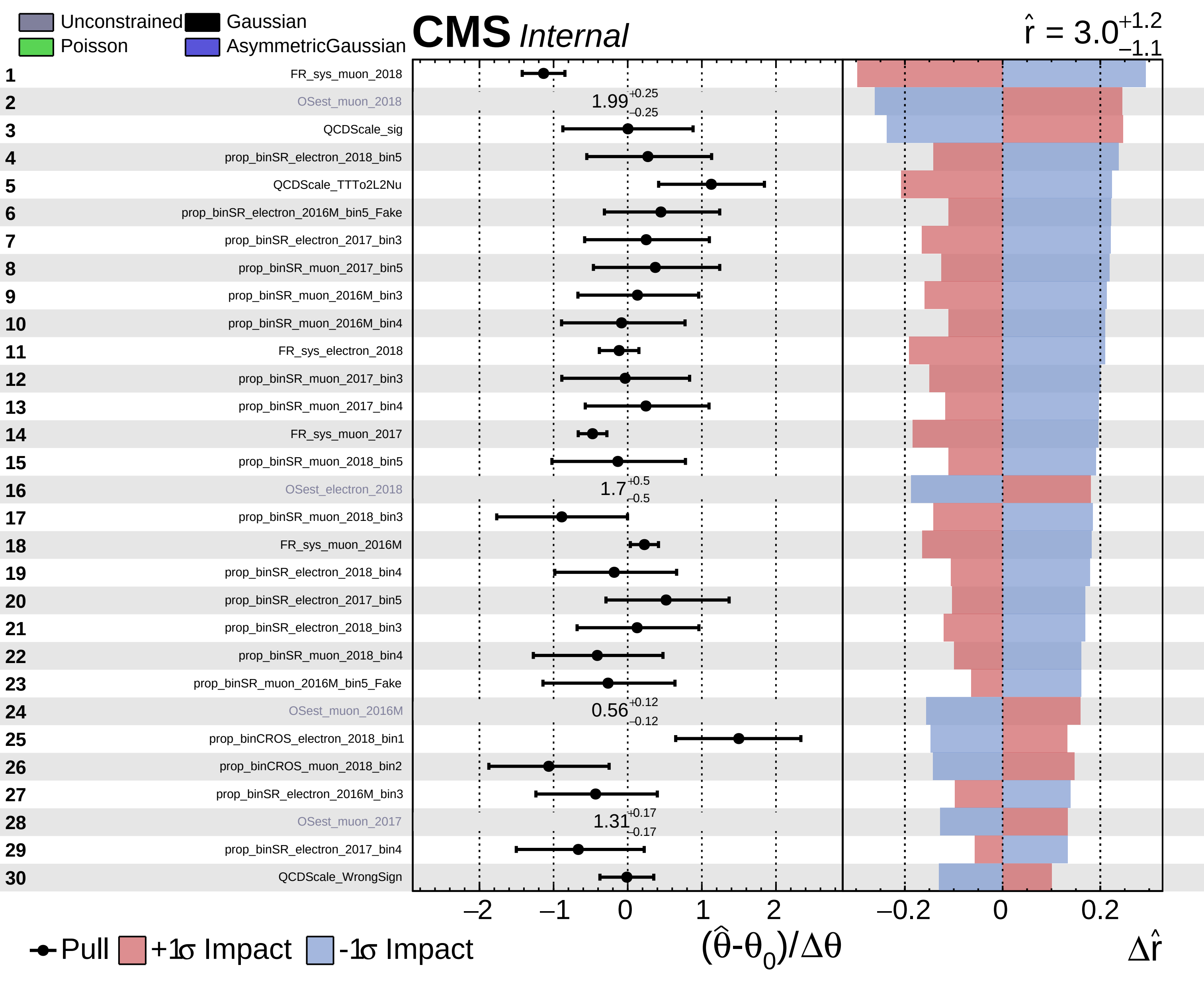
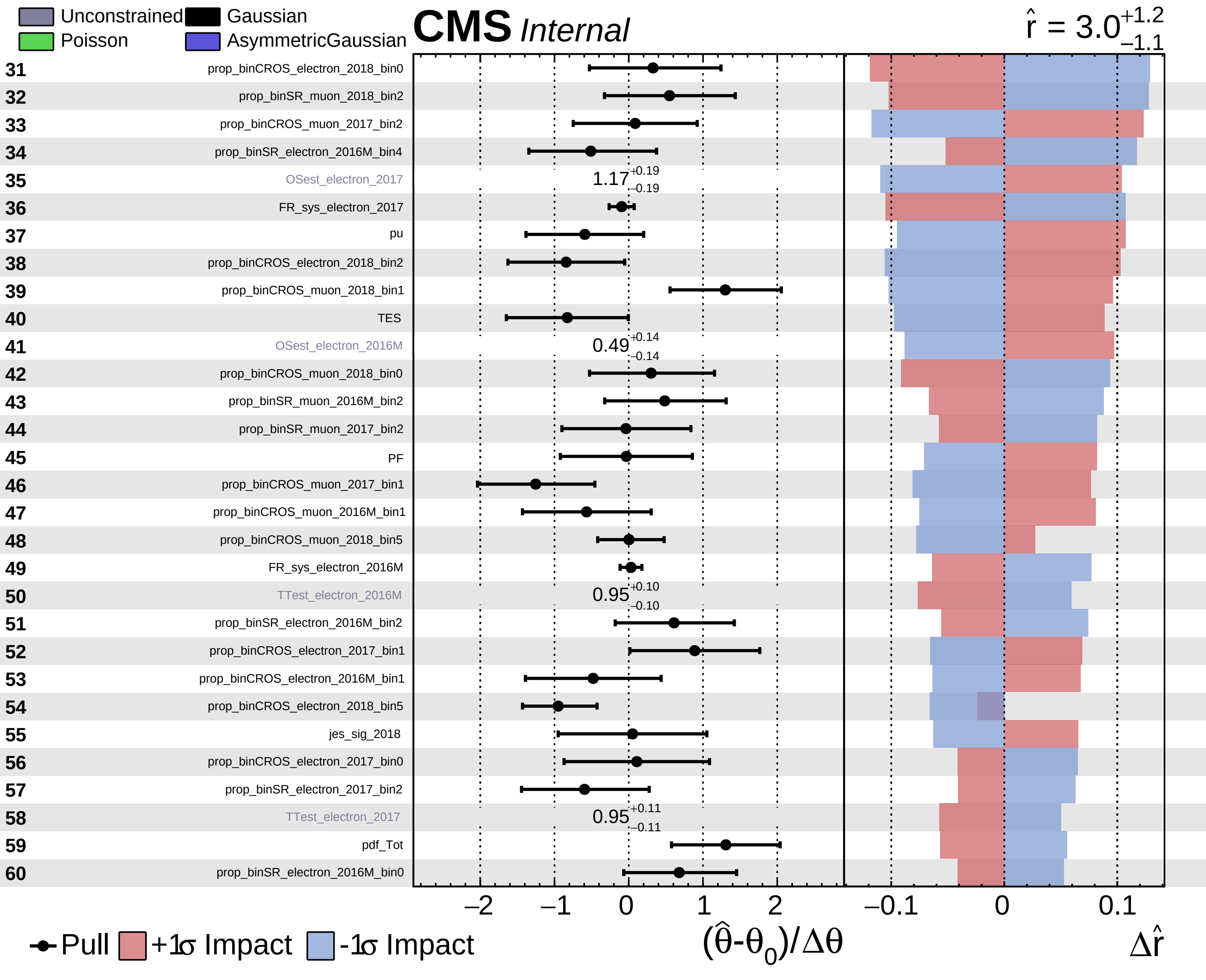
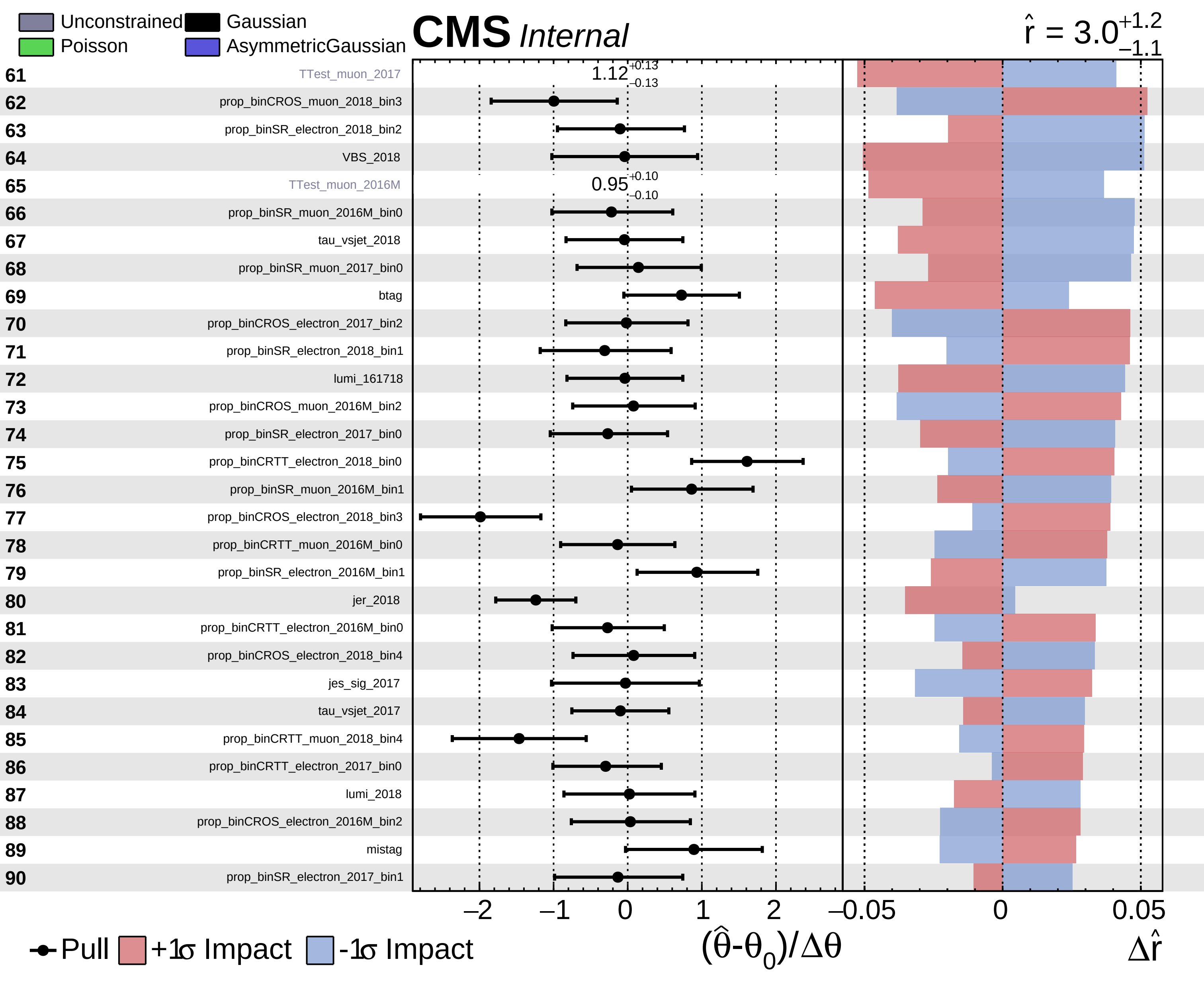


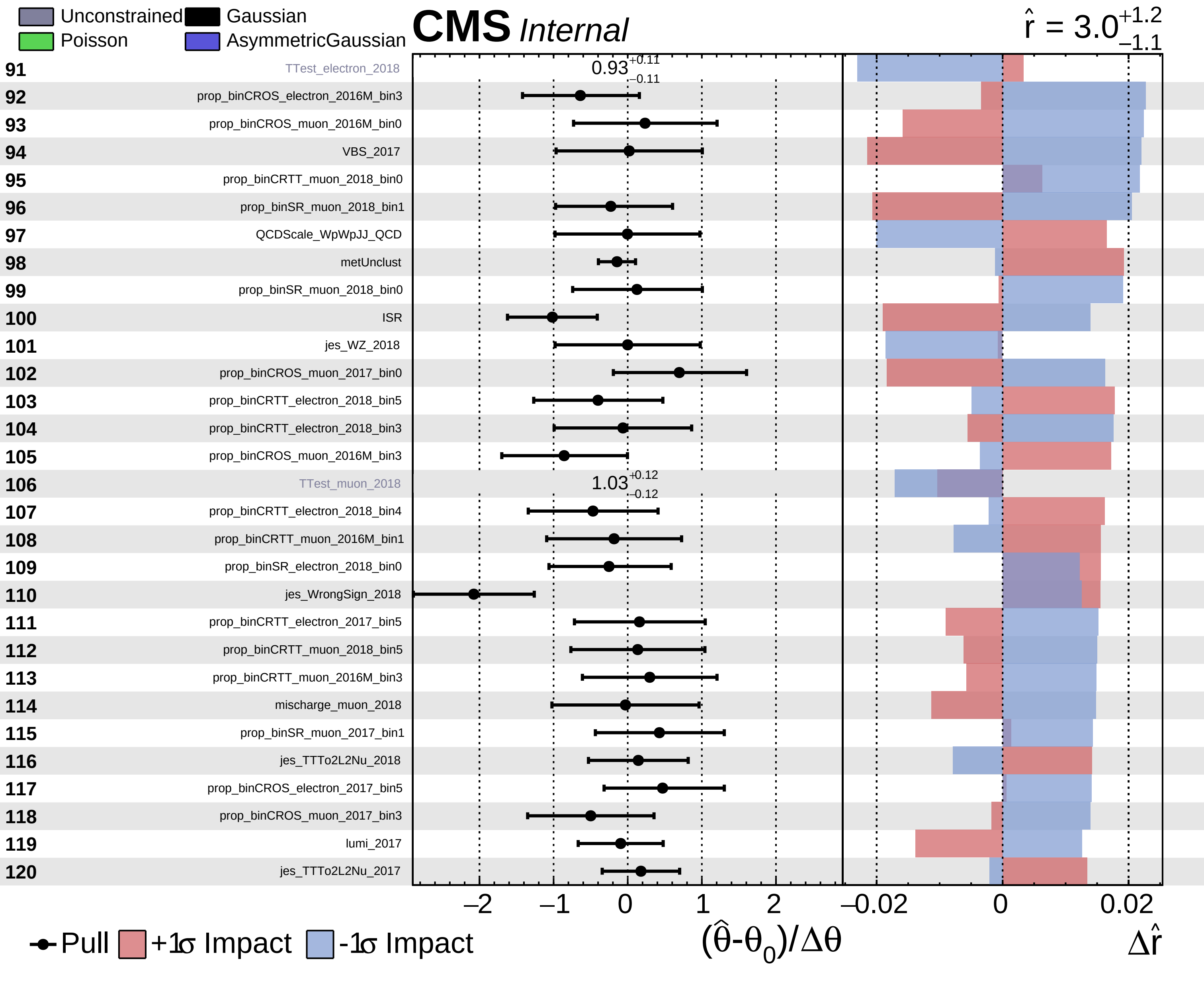
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 3.0^{+1.2}_{-1.1}$
1 FR_sys_muon_2018
2 OSest_muon_2018 $1.99^{+0.25}_{-0.25}$
3 QCDScale_sig
4 prop_binSR_electron_2018_bin5
5 QCDScale_TTTo2L2Nu
6 prop_binSR_electron_2016M_bin5_Fake
7 prop_binSR_electron_2017_bin3
8 prop_binSR_muon_2017_bin5
9 prop_binSR_muon_2016M_bin3
10 prop_binSR_muon_2016M_bin4
11 FR_sys_electron_2018
12 prop_binSR_muon_2017_bin3
13 prop_binSR_muon_2017_bin4
14 FR_sys_muon_2017
15 prop_binSR_muon_2018_bin5
16 OSest_electron_2018 $1.7^{+0.5}_{-0.5}$
17 prop_binSR_muon_2018_bin3
18 FR_sys_muon_2016M
19 prop_binSR_electron_2018_bin4
20 prop_binSR_electron_2017_bin5
21 prop_binSR_electron_2018_bin3
22 prop_binSR_muon_2018_bin4
23 prop_binSR_muon_2016M_bin5_Fake
24 OSest_muon_2016M $0.56^{+0.12}_{-0.12}$
25 prop_binCROS_electron_2018_bin1
26 prop_binCROS_muon_2018_bin2
27 prop_binSR_electron_2016M_bin3
28 OSest_muon_2017 $1.31^{+0.17}_{-0.17}$
29 prop_binSR_electron_2017_bin4
30 QCDScale_WrongSign
−2 −1 0 1 2
−0.2 0 0.2
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



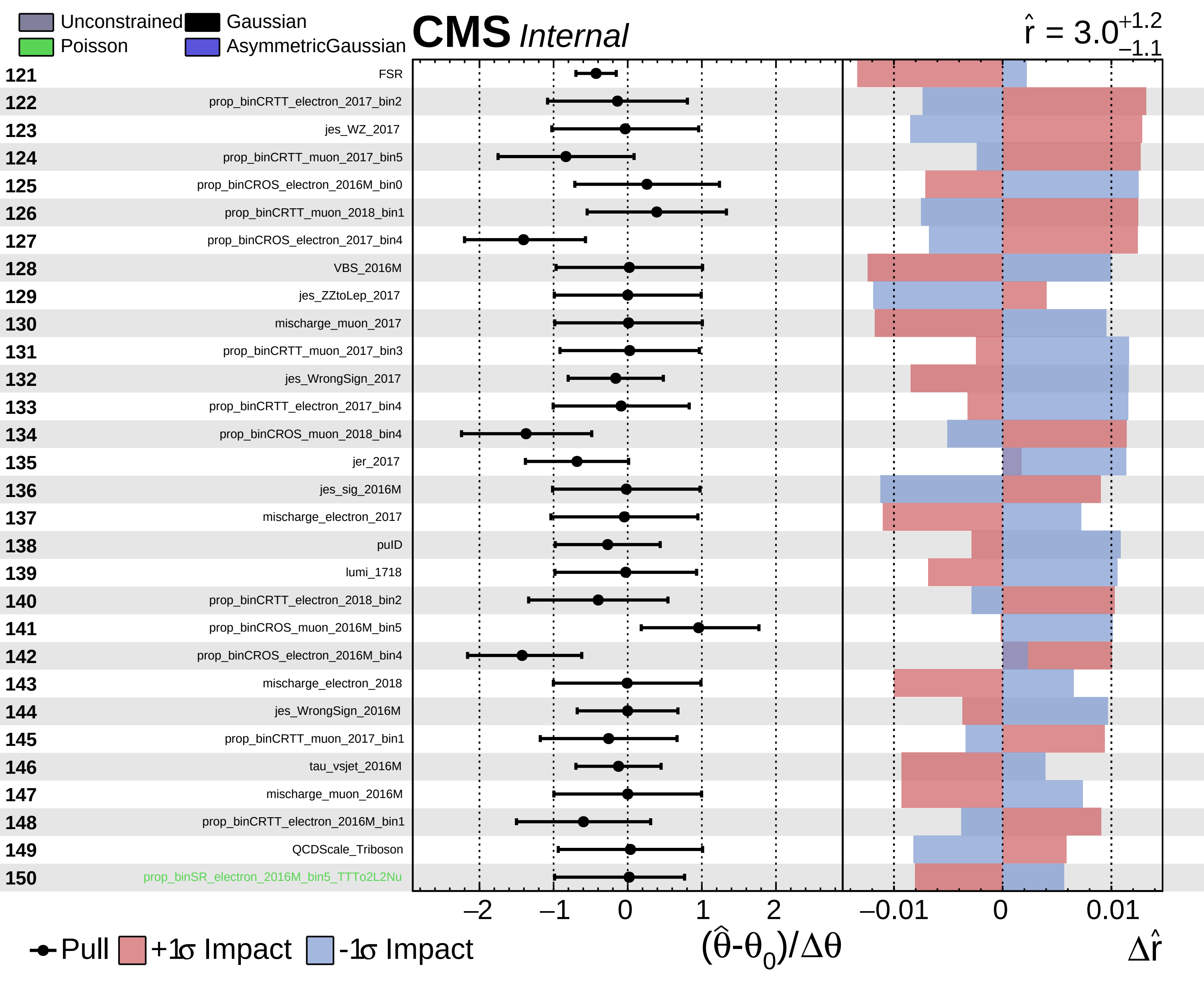
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 3.0^{+1.2}_{-1.1}$
31 prop_binCROS_electron_2018_bin0
32 prop_binSR_muon_2018_bin2
33 prop_binCROS_muon_2017_bin2
34 prop_binSR_electron_2016M_bin4
35 OSest_electron_2017 $1.17^{+0.19}_{-0.19}$
36 FR_sys_electron_2017
37 pu
38 prop_binCROS_electron_2018_bin2
39 prop_binCROS_muon_2018_bin1
40 TES
41 OSest_electron_2016M $0.49^{+0.14}_{-0.14}$
42 prop_binCROS_muon_2018_bin0
43 prop_binSR_muon_2016M_bin2
44 prop_binSR_muon_2017_bin2
45 PF
46 prop_binCROS_muon_2017_bin1
47 prop_binCROS_muon_2016M_bin1
48 prop_binCROS_muon_2018_bin5
49 FR_sys_electron_2016M
50 TTest_electron_2016M $0.95^{+0.10}_{-0.10}$
51 prop_binSR_electron_2016M_bin2
52 prop_binCROS_electron_2017_bin1
53 prop_binCROS_electron_2016M_bin1
54 prop_binCROS_electron_2018_bin5
55 jes_sig_2018
56 prop_binCROS_electron_2017_bin0
57 prop_binSR_electron_2017_bin2
58 TTest_electron_2017 $0.95^{+0.11}_{-0.11}$
59 pdf_Tot
60 prop_binSR_electron_2016M_bin0
−2 −1 0 1 2
−0.1 0 0.1
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



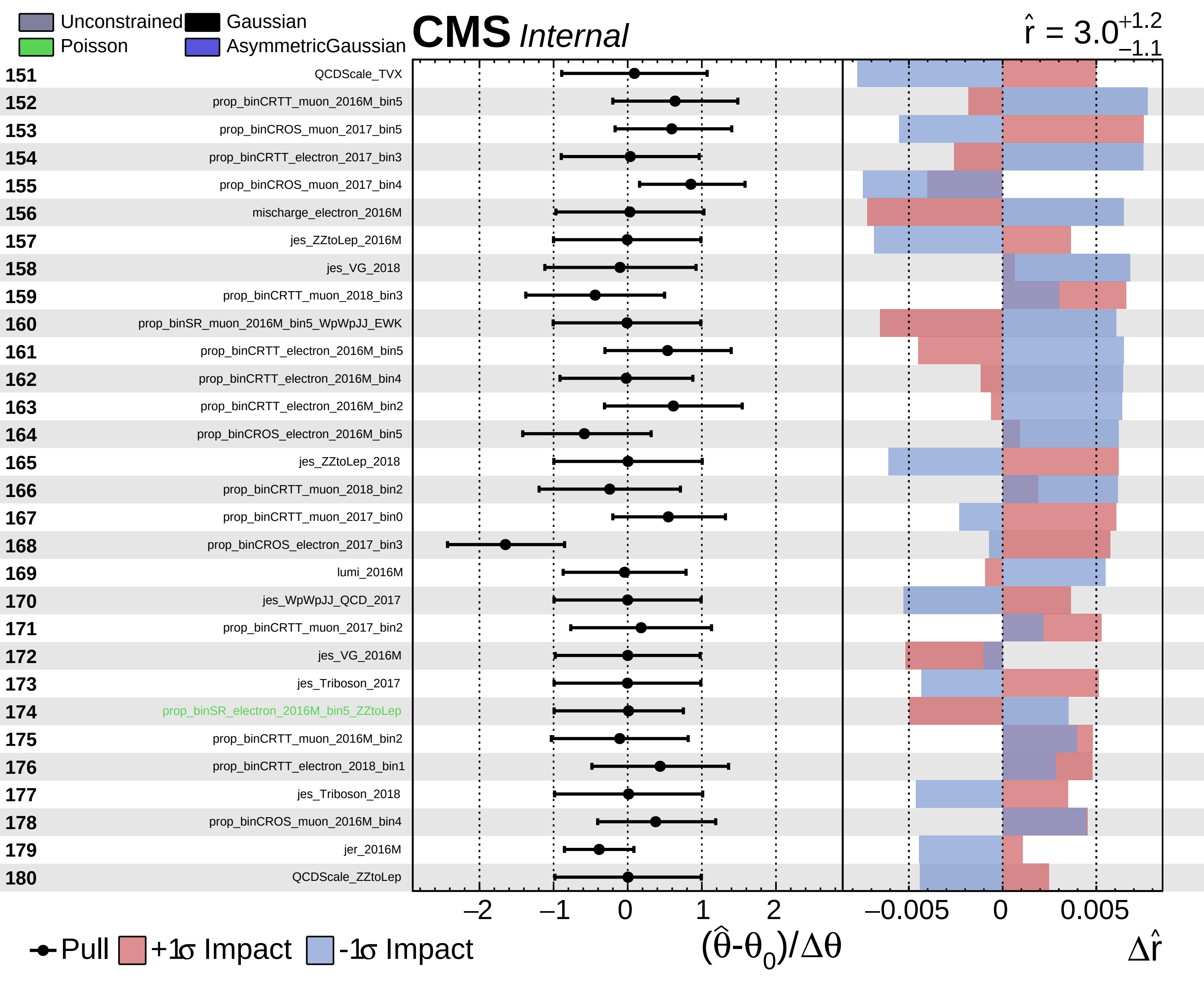
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 3.0^{+1.2}_{-1.1}$
61 TTest_muon_2017 $1.12^{+0.13}_{-0.13}$
62 prop_binCROS_muon_2018_bin3
63 prop_binSR_electron_2018_bin2
64 VBS_2018
65 TTest_muon_2016M $0.95^{+0.10}_{-0.10}$
66 prop_binSR_muon_2016M_bin0
67 tau_vsjet_2018
68 prop_binSR_muon_2017_bin0
69 btag
70 prop_binCROS_electron_2017_bin2
71 prop_binSR_electron_2018_bin1
72 lumi_161718
73 prop_binCROS_muon_2016M_bin2
74 prop_binSR_electron_2017_bin0
75 prop_binCRTT_electron_2018_bin0
76 prop_binSR_muon_2016M_bin1
77 prop_binCROS_electron_2018_bin3
78 prop_binCRTT_muon_2016M_bin0
79 prop_binSR_electron_2016M_bin1
80 jer_2018
81 prop_binCRTT_electron_2016M_bin0
82 prop_binCROS_electron_2018_bin4
83 jes_sig_2017
84 tau_vsjet_2017
85 prop_binCRTT_muon_2018_bin4
86 prop_binCRTT_electron_2017_bin0
87 lumi_2018
88 prop_binCROS_electron_2016M_bin2
89 mistag
90 prop_binSR_electron_2017_bin1
−2 −1 0 1 2
−0.05 0 0.05
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



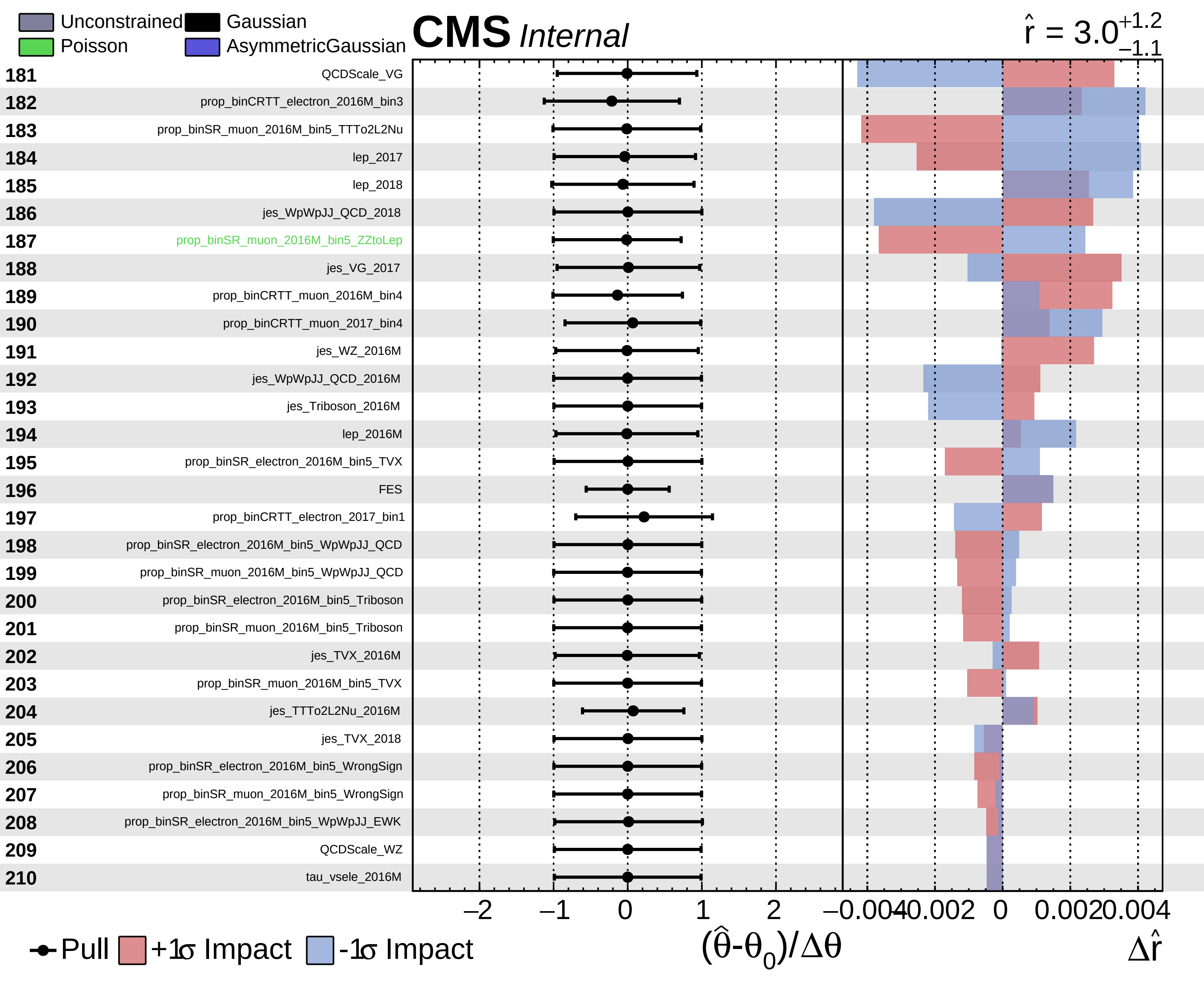
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 3.0^{+1.2}_{-1.1}$
91 TTest_electron_2018
$0.93^{+0.11}_{-0.11}$
92 prop_binCROS_electron_2016M_bin3
93 prop_binCROS_muon_2016M_bin0
94 VBS_2017
95 prop_binCRTT_muon_2018_bin0
96 prop_binSR_muon_2018_bin1
97 QCDScale_WpWpJJ_QCD
98 metUnclust
99 prop_binSR_muon_2018_bin0
100 ISR
101 jes_WZ_2018
102 prop_binCROS_muon_2017_bin0
103 prop_binCRTT_electron_2018_bin5
104 prop_binCRTT_electron_2018_bin3
105 prop_binCROS_muon_2016M_bin3
106 TTest_muon_2018
$1.03^{+0.12}_{-0.12}$
107 prop_binCRTT_electron_2018_bin4
108 prop_binCRTT_muon_2016M_bin1
109 prop_binSR_electron_2018_bin0
110 jes_WrongSign_2018
111 prop_binCRTT_electron_2017_bin5
112 prop_binCRTT_muon_2018_bin5
113 prop_binCRTT_muon_2016M_bin3
114 mischarge_muon_2018
115 prop_binSR_muon_2017_bin1
116 jes_TTTo2L2Nu_2018
117 prop_binCROS_electron_2017_bin5
118 prop_binCROS_muon_2017_bin3
119 lumi_2017
120 jes_TTTo2L2Nu_2017
−2 −1 0 1 2
−0.02 0 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



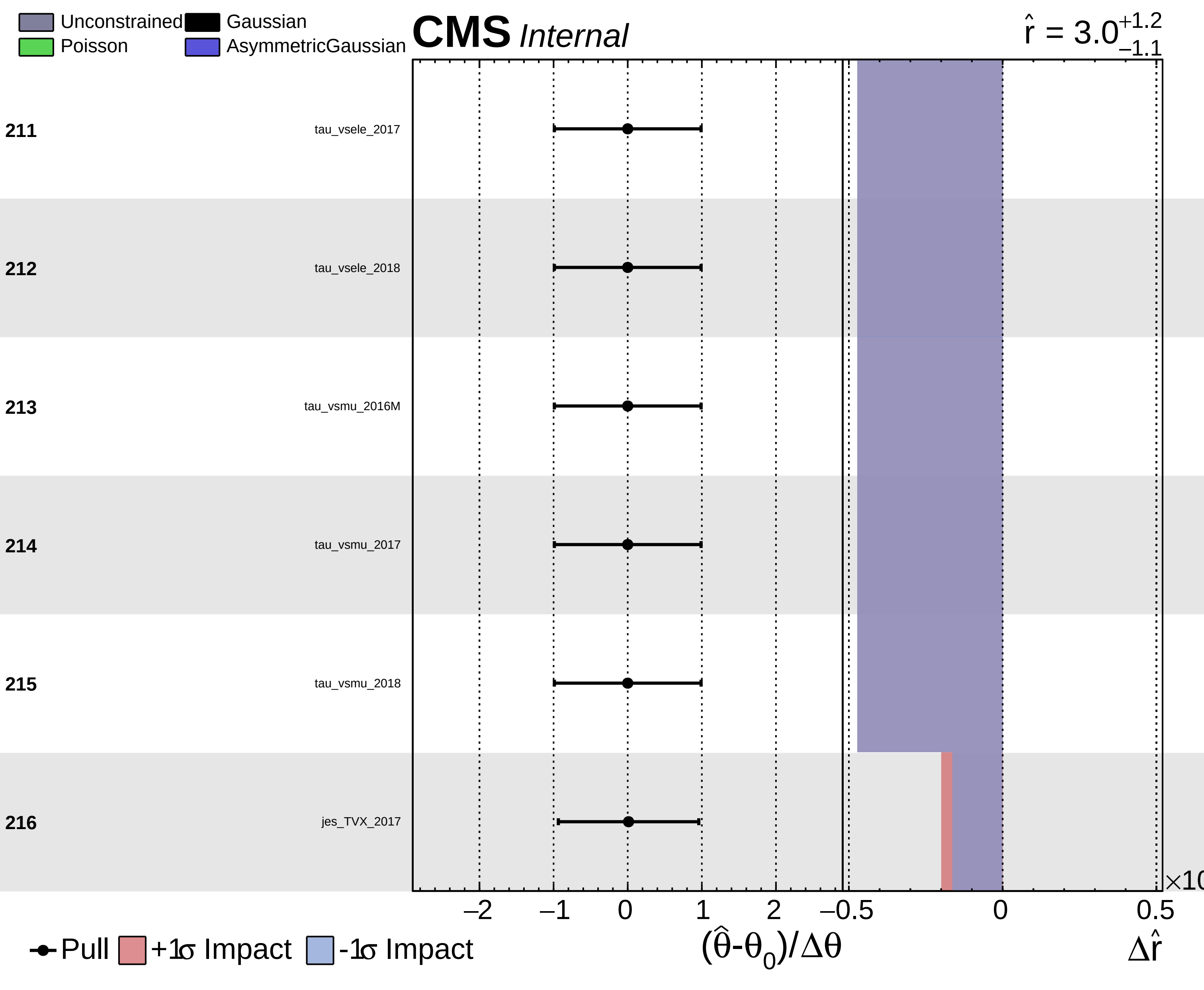
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 3.0^{+1.2}_{-1.1}$
121 FSR
122 prop_binCRTT_electron_2017_bin2
123 jes_WZ_2017
124 prop_binCRTT_muon_2017_bin5
125 prop_binCROS_electron_2016M_bin0
126 prop_binCRTT_muon_2018_bin1
127 prop_binCROS_electron_2017_bin4
128 VBS_2016M
129 jes_ZZtoLep_2017
130 mischarge_muon_2017
131 prop_binCRTT_muon_2017_bin3
132 jes_WrongSign_2017
133 prop_binCRTT_electron_2017_bin4
134 prop_binCROS_muon_2018_bin4
135 jer_2017
136 jes_sig_2016M
137 mischarge_electron_2017
138 puID
139 lumi_1718
140 prop_binCRTT_electron_2018_bin2
141 prop_binCROS_muon_2016M_bin5
142 prop_binCROS_electron_2016M_bin4
143 mischarge_electron_2018
144 jes_WrongSign_2016M
145 prop_binCRTT_muon_2017_bin1
146 tau_vsjet_2016M
147 mischarge_muon_2016M
148 prop_binCRTT_electron_2016M_bin1
149 QCDScale_Triboson
150 prop_binSR_electron_2016M_bin5_TTTo2L2Nu
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 3.0^{+1.2}_{-1.1}$
151 QCDScale_TVX
152 prop_binCRTT_muon_2016M_bin5
153 prop_binCROS_muon_2017_bin5
154 prop_binCRTT_electron_2017_bin3
155 prop_binCROS_muon_2017_bin4
156 mischarge_electron_2016M
157 jes_ZZtoLep_2016M
158 jes_VG_2018
159 prop_binCRTT_muon_2018_bin3
160 prop_binSR_muon_2016M_bin5_WpWpJJ_EWK
161 prop_binCRTT_electron_2016M_bin5
162 prop_binCRTT_electron_2016M_bin4
163 prop_binCRTT_electron_2016M_bin2
164 prop_binCROS_electron_2016M_bin5
165 jes_ZZtoLep_2018
166 prop_binCRTT_muon_2018_bin2
167 prop_binCRTT_muon_2017_bin0
168 prop_binCROS_electron_2017_bin3
169 lumi_2016M
170 jes_WpWpJJ_QCD_2017
171 prop_binCRTT_muon_2017_bin2
172 jes_VG_2016M
173 jes_Triboson_2017
174 prop_binSR_electron_2016M_bin5_ZZtoLep
175 prop_binCRTT_muon_2016M_bin2
176 prop_binCRTT_electron_2018_bin1
177 jes_Triboson_2018
178 prop_binCROS_muon_2016M_bin4
179 jer_2016M
180 QCDScale_ZZtoLep
−2 −1 0 1 2
−0.005 0 0.005
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 3.0^{+1.2}_{-1.1}$
181 QCDScale_VG
182 prop_binCRTT_electron_2016M_bin3
183 prop_binSR_muon_2016M_bin5_TTTo2L2Nu
184 lep_2017
185 lep_2018
186 jes_WpWpJJ_QCD_2018
187 prop_binSR_muon_2016M_bin5_ZZtoLep
188 jes_VG_2017
189 prop_binCRTT_muon_2016M_bin4
190 prop_binCRTT_muon_2017_bin4
191 jes_WZ_2016M
192 jes_WpWpJJ_QCD_2016M
193 jes_Triboson_2016M
194 lep_2016M
195 prop_binSR_electron_2016M_bin5_TVX
196 FES
197 prop_binCRTT_electron_2017_bin1
198 prop_binSR_electron_2016M_bin5_WpWpJJ_QCD
199 prop_binSR_muon_2016M_bin5_WpWpJJ_QCD
200 prop_binSR_electron_2016M_bin5_Triboson
201 prop_binSR_muon_2016M_bin5_Triboson
202 jes_TVX_2016M
203 prop_binSR_muon_2016M_bin5_TVX
204 jes_TTTo2L2Nu_2016M
205 jes_TVX_2018
206 prop_binSR_electron_2016M_bin5_WrongSign
207 prop_binSR_muon_2016M_bin5_WrongSign
208 prop_binSR_electron_2016M_bin5_WpWpJJ_EWK
209 QCDScale_WZ
210 tau_vsele_2016M
−2
−1
0
1
2
−0.004
−0.002
0
0.002
0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 3.0^{+1.2}_{-1.1}$
211
tau_vsele_2017
212
tau_vsele_2018
213
tau_vsmu_2016M
214
tau_vsmu_2017
215
tau_vsmu_2018
216
jes_TVX_2017
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$